



A smart pizza box?

Seen in India and Dubai, the VenTiT box is considered to be the world's most advanced pizza box. Read more: <http://dgit.in/Smartyzza>



Breaking Good

Check out this YouTube video that re-imagines Breaking Bad with happier characters. <http://dgit.in/BreakGood>

```
NULL AUTO_INCREMENT,
  `longurl` text NOT NULL,
  PRIMARY KEY (`id`)
) ENGINE=InnoDB DEFAULT
CHARSET=utf8 AUTO_INCRE-
MENT=16 ;
```

Now comes the home page of our site, as we are creating just a demo so it need not be fancy. We just need to display a text input field (where the user will put the long URL) and a submit button. Save this file as "index.html"

index.html code

```
<form action="process.php"
method="post"> <!--We speci-
fy the file where this form
would be processed and we'll
use the post method-->
URL(with "http://"): <input
type="text" name="longurl">
<input type="submit">
```

Now the form(for home page) is created, we now need to create the "process.php" file. This file is responsible to add the long URL in the DB and then 'echo' the short URL with its ID.

process.php code

```
<?php
//Make connection with database so that
we can put the long url in it.
$con=mysqli_connect("localhost","root",
"pass_if_any","urldb") or
die("Connection Error");
//Assign the variable $longurl to the
user's long address
$longurl = $_POST['longurl'];
```

```
//Put the longURL in the DB
$sql=mysqli_query($con,"INSERT INTO
urlltable (longurl)VALUES('$longurl')");
//Now we selects the ID to which this
long URL has been assigned.
$result = mysqli_
query($con,"SELECT id FROM
urlltable WHERE
longurl='$longurl'");
while($id = mysqli_fetch_
array($result))
{
  echo "URL Shortened Suc-
cessfully! \".'<br>'. "Your
shortened URL is "; //Dis-
play the message
  echo 'http://localhost/
shortener/' . $id['id']; //
and finally display the
short URL with ID
}
mysqli_close($con);
?>
```

So we have put the long URL in the data- base and the 50% of the task is finished. Now we just need to code the redirection method for which we need to edit the ".htaccess" file, it is responsible to parse the ID from URL, via redirect.php file thus making the redirection possible. Save the following code as ".htaccess"; don't forget there's a full-stop in the name of this file before 'htaccess'.

.htaccess code

```
*Turn on rewrite rules in Apache server
RewriteEngine On
*It tells how to handle URLs
RewriteRule ^([0-9]+)$ redirect.php?l=$1
```

This is the most important file as it is responsible for all the redirection taking places. This file basically takes the "ID" from the URL, then it will query the DB for the corresponding longurl and then the user is redirected or a error message is displayed if the ID is invalid. Remember to save this file with name as "redirect.php".

redirect.php code

```
<?php
//check whether the ID is provided or not,
if not then condition won't be executed
```

```
$LocationId = (isset($_GET['l'])) ? $_
GET['l'] : null;
if( $LocationId )
{ //Make connection to DB
  $DBConn = mysqli_connect("localhost
","root","pass_if_any","urldb") or
die("Connection Error");
//Selects the URL which corresponding
to the ID as given by the user in URL
  $Resut = mysqli_query($DBConn,
sprintf("SELECT longurl FROM urlla-
ble WHERE id LIKE '%s' LIMIT 0,1",
mysqli_real_escape_string($DBConn,
$LocationId))) or die( mysqli_error() );
  if( $Resut )
  { $Row = mysqli_fetch_
array($Resut);
    if( isset($Row['longurl']) &&
!empty($Row['longurl']) ) //checks
whether URL exist or not
    { header('HTTP/1.1 301 Moved
Permanently'); //if url exist in DB, then
set the header to 301 and redirect user to
the longurl
      header(sprintf('Location: %s',
$Row['longurl'])); //Redirects the user to
the long URL
      exit;
    }else {echo "Invalid URL"; //If inva-
lid ID given by user then return this
error}}}?>
```

So now everything is done! Make sure to keep all the files (include .htaccess) in the same folder. To open the site go to the address that is linked to your Apache server. This whole script is writ- ten by me and is just a short presenta- tion to show you all the basics of creat- ing a URL shortener. Currently there are many vulnerabilities in this script. Such as this script can get hacked easily as there are XSS vulnerabilities. You can add CSS to make the site look more fancy and then you can also add features such as 'hit' counter and other different things. Possibilities are unend- ing with PHP, it just depends on how you make use of your skills to bring out the best in it.

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